

Chapter 2

SYSTEM ANALYSIS

2.1 Introduction

System analysis is a vital phase in software development that focuses on studying existing practices, identifying their drawbacks, and defining the requirements for an improved system. In healthcare, especially in cancer screening and oncology management, system analysis helps uncover inefficiencies related to diagnosis delays, workflow fragmentation, and poor data coordination. For oral cancer detection and treatment, early diagnosis and seamless coordination between doctors, laboratories, oncologists, and patients are critical. This chapter analyzes the current oral cancer screening and management systems, identifies their limitations, and presents the proposed ML-powered OraTrack system along with its feasibility and development methodology. It also examines the practical, operational, and economic aspects of implementing a digital, AI-assisted platform. By evaluating current workflows, data handling methods, and communication channels, system analysis highlights the specific areas where automation and machine learning can improve efficiency. Moreover, it provides a framework for designing a patient-centric, transparent, and reliable system that can enhance treatment outcomes and reduce delays in oral cancer diagnosis and management. This analysis forms the foundation for developing a robust platform that integrates predictive analytics, real-time monitoring, and structured oncology care within a single, unified interface.

2.2 Existing System

The existing oral cancer detection and oncology management system is largely manual and fragmented. Most hospitals rely on traditional clinical examination methods, where doctors assess oral lesions visually and recommend further tests based on experience. Patient records, screening details, biopsy requests, lab reports, and treatment histories are often maintained in paper files or isolated digital systems. Communication between doctors, laboratories, oncologists, and patients is slow and unstructured. Biopsy workflows lack transparency, and patients are frequently unaware of the current status of their test results. Oncology treatment management is also handled separately, leading to duplication of data, delayed follow-ups, and inefficient monitoring of patient symptoms and therapy responses.

2.2.1 Limitations of Existing System

- No intelligent or ML-based cancer risk prediction system
- Heavy dependence on manual clinical judgment
- Delayed diagnosis due to slow biopsy and reporting workflows
- Paper-based or fragmented patient records
- Lack of transparency in biopsy status and lab processing

- Poor coordination between doctors, lab technicians, and oncologists
- No real-time patient symptom monitoring after treatment
- Limited patient–doctor communication channels
- Absence of emergency alert mechanisms for critical conditions
- High risk of data loss, miscommunication, and delayed treatment

2.3 Proposed System

The proposed system, OraTrack, is an ML-powered Oral Cancer Detection and Smart Oncology Patient Management System designed to address the limitations of traditional healthcare workflows. The system uses a pre-trained Random Forest machine learning model to predict oral cancer risk based solely on doctor-entered clinical parameters such as lesion size, ulcer duration, pain intensity, bleeding, lifestyle habits, and medical history. Based on the predicted risk level, the system automatically initiates a biopsy request for moderate or high-risk cases.

OraTrack integrates a biopsy tracking system that allows lab technicians to update biopsy status at each stage, ensuring transparency and timely communication. Once cancer is confirmed, the system automatically assigns an oncologist and transfers all patient data for continued treatment management. The platform supports comprehensive oncology workflows, including treatment planning, therapy scheduling, symptom monitoring, emergency alerts, and two-way communication between patients and doctors. Administrators oversee the entire system, ensuring security, analytics, and smooth operation.

2.3.1 Advantages of Proposed System

- ML-based accurate oral cancer risk prediction
- Early detection and faster clinical decision-making
- Automated biopsy request and tracking system
- Transparent and time-efficient lab workflow
- Seamless transfer from screening to oncology treatment
- Centralized digital patient records
- Continuous symptom and treatment monitoring
- Two-way patient–doctor communication
- Emergency alert support for critical situations
- Improved coordination among doctors, labs, and oncologists
- Reduced delays, errors, and paperwork
- Scalable and intelligent healthcare management system

2.4 Feasibility Study

A feasibility study evaluates the practicality of implementing the proposed system from technical, operational, and economic perspectives. It helps determine whether the system can be developed successfully using available resources, technologies, and constraints. Conducting a feasibility study ensures that the project is realistic, cost-effective, and capable of meeting user requirements. It also helps identify potential risks and challenges at an early stage of development. For a healthcare-oriented system like OraTrack, feasibility analysis is essential to ensure accuracy, reliability, security, and ease of operation. The study assesses whether the proposed solution can integrate smoothly into existing medical workflows. It further examines the system's ability to handle real-time data, multiple user roles, and sensitive patient information. By analyzing feasibility, the project ensures optimal utilization of technical resources, operational efficiency, and economic sustainability. Overall, the feasibility study confirms that the proposed system is practical, beneficial, and suitable for real-world deployment in medical institutions.

2.4.1 Technical Feasibility

The system is technically feasible as it is developed using reliable and widely adopted technologies such as Python, Django, MySQL, and machine learning libraries like scikit-learn. The Random Forest algorithm provides stable and accurate predictions. Role-based access, and automated workflows can be effectively implemented within the chosen technology stack.

2.4.2 Operational Feasibility

OraTrack is operationally feasible due to its user-friendly interface and role-specific modules for administrators, doctors, lab technicians, oncologists, and patients. Automation reduces manual workload, while real-time notifications and dashboards enhance usability. Minimal training is required for medical staff to operate the system effectively.

2.4.3 Economic Feasibility

The system is economically feasible as it uses open-source technologies, minimizing development and licensing costs. Digital record management reduces paperwork expenses and operational delays. Early diagnosis and efficient treatment management can significantly reduce long-term healthcare costs for hospitals and patients.

2.5 Software Engineering Paradigm Applied

The proposed system is developed using the Agile software engineering paradigm, which emphasizes flexibility, iterative development, and continuous improvement. Agile divides the overall development process into smaller development cycles known as sprints, allowing the system to be built, tested, and refined incrementally. Each sprint involves requirement analysis, system design, implementation, testing, and review, ensuring that individual modules are functional and reliable before proceeding to subsequent iterations. This approach enables early identification of errors and reduces the risk of major system failures.

Although Agile encourages continuous user and stakeholder feedback, direct interaction with real users is limited in the academic environment. Therefore, feedback is incorporated through system testing results, predefined healthcare workflows, performance evaluation, and periodic academic reviews. These evaluations guide improvements to system features such as screening workflows, biopsy tracking, oncology case management, and emergency alert mechanisms. The Agile paradigm also supports rapid adaptation to evolving requirements, enabling the integration of additional medical parameters, improved machine learning models, and enhanced security mechanisms.

Continuous testing and incremental integration improve system performance, reliability, and data accuracy. Overall, the Agile software engineering paradigm ensures structured development, improved software quality, reduced development risk, and effective implementation of a robust, intelligent, and scalable healthcare management system suitable for academic evaluation.

